

Template

Studentnames and studentnumbers here

2026-06-15

Set-up your environment

```
require(tidyverse)
```

```
## Carregando pacotes exigidos: tidyverse
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
require(rmarkdown)
```

```
## Carregando pacotes exigidos: rmarkdown
```

```
require(yaml)
```

```
## Carregando pacotes exigidos: yaml
```

```
library(readxl)
```

```
library(sf)
```

```
## Linking to GEOS 3.11.2, GDAL 3.8.2, PROJ 9.3.1; sf_use_s2() is TRUE
```

```
library(giscoR)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(tinytex)
```

```
tinytex::install_tinytex(force = TRUE)
```

```
## tlmgr install tlgpg
## tlmgr update --self
## tlmgr install tlgpg
## tlmgr --repository https://www.preining.info/tlgpg/ install tlgpg
## tlmgr option repository "https://tlnet.yihui.org"
## tlmgr update --list
```

Title Page

Include your names Noel Kona Giorgia Tanzi Iann Soffried Julian Trinkl Morris Nouwens Dania Al Khulaifi

Include the tutorial group number Tutorial 5 Group 5

Include your tutorial lecturer's name Simon Loop

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant?
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
rent<-read_xlsx("prc_hicp_aind__custom_12680989_spreadsheet (1).xlsx")
```

```
## New names:
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
```

```
## * '' -> '...14'
## * '' -> '...15'
## * '' -> '...16'
## * '' -> '...17'
## * '' -> '...18'
## * '' -> '...19'
## * '' -> '...20'
## * '' -> '...21'
## * '' -> '...22'
## * '' -> '...23'
## * '' -> '...24'
## * '' -> '...25'
## * '' -> '...26'
## * '' -> '...27'
## * '' -> '...28'
## * '' -> '...29'
## * '' -> '...30'
## * '' -> '...31'
## * '' -> '...32'
## * '' -> '...33'
```

```
europe <- gisco_get_countries(
  resolution = "20"
)

wage<-read_xlsx("tps00155_page_spreadsheet.xlsx")
```

```
## New names:
## * '' -> '...2'
## * '' -> '...3'
## * '' -> '...4'
## * '' -> '...5'
## * '' -> '...6'
## * '' -> '...7'
## * '' -> '...8'
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
## * '' -> '...12'
## * '' -> '...13'
## * '' -> '...14'
## * '' -> '...15'
## * '' -> '...16'
## * '' -> '...17'
## * '' -> '...18'
## * '' -> '...19'
## * '' -> '...20'
## * '' -> '...21'
## * '' -> '...22'
## * '' -> '...23'
## * '' -> '...24'
## * '' -> '...25'
```

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
# =====
#rent cleaning, deleted NA columns and only selected values after 2014 and then calculated var between
# =====
colnames(rent)<-rent[8,]

rent<-rent[-1:-10,]

rent<-rent[-36:-42,]

rent<-rent[, -seq(from=3, to=ncol(rent), by=2)]

rent<-rent[-33,]

rent<-rent[, -2:-6]

rent<-rent[-34,]

rent <- rent %>%
  mutate(
    `2015` = as.numeric(gsub("[^0-9.]", "", `2015`)),
    `2025` = as.numeric(gsub("[^0-9.]", "", `2025`)),
    Var = (`2025` - `2015`) / `2015`
  )

rent <- rent %>%
```

```

mutate(across(2:ncol(rent), as.numeric))

# =====
# europe cleaning
# =====

europe <- europe[europe$CNTR_ID %in% c(
  "NL", "DE", "FR", "BE", "IT", "ES", "PT",
  "AT", "PL", "CZ", "SK", "HU", "RO",
  "BG", "HR", "SI", "DK", "SE", "FI",
  "EE", "LV", "LT", "IE", "LU", "GR",
  "CY", "MT", "NO", "CH", "ME", "MK", "TR", "IS", "RS"
), ]

# =====
# wage cleaning
# =====

colnames(wage)<-wage[7,]

wage<-wage[c(-1:-12,-16,-24:-25,-32,-38:-44,-53:-57),]

wage<-wage[,-seq(from=3, to=ncol(wage), by=2)]

wage<-wage[,-seq(from=3, to=ncol(wage), by=2)]

```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

```
data<-left_join(europe, rent, join_by("NAME_ENGL"=="TIME"))
```

Variable 2

```

hicp_long <- rent %>%
  pivot_longer(
    cols = matches("^20"),
    names_to = "year",
    values_to = "hicp"
  ) %>%
  mutate(year = as.numeric(year))

wage_long <- wage %>%
  pivot_longer(
    cols = matches("^20"),
    names_to = "year",
    values_to = "wage_growth"
  ) %>%

```

```

mutate(
  year = as.numeric(str_extract(year, "\\d{4}")),
  wage_growth = as.numeric(wage_growth)
)
analysis <- left_join(
  hicp_long,
  wage_long,
  by = c("TIME", "year")
)

```

3.3 Visualize temporal variation

```

# =====
# Calculate EU average each year
# =====
temporal <- analysis %>%
  group_by(year) %>%
  summarise(
    avg_hicp = mean(hicp, na.rm = TRUE),
    avg_wage = mean(wage_growth, na.rm = TRUE)
  ) %>%
  arrange(year) %>%
  mutate(
    avg_hicp      = avg_hicp / avg_hicp[year == 2020] * 100,
    avg_wage_growth = avg_wage / avg_wage[year == 2020] * 100
  ) %>%
  select(year, avg_hicp, avg_wage_growth)

# =====
# Convert to long format for plotting
# =====
temporal_long <- temporal %>%
  filter(year >= 2020) %>%
  pivot_longer(
    cols = c(avg_hicp, avg_wage_growth),
    names_to = "variable",
    values_to = "value"
  )

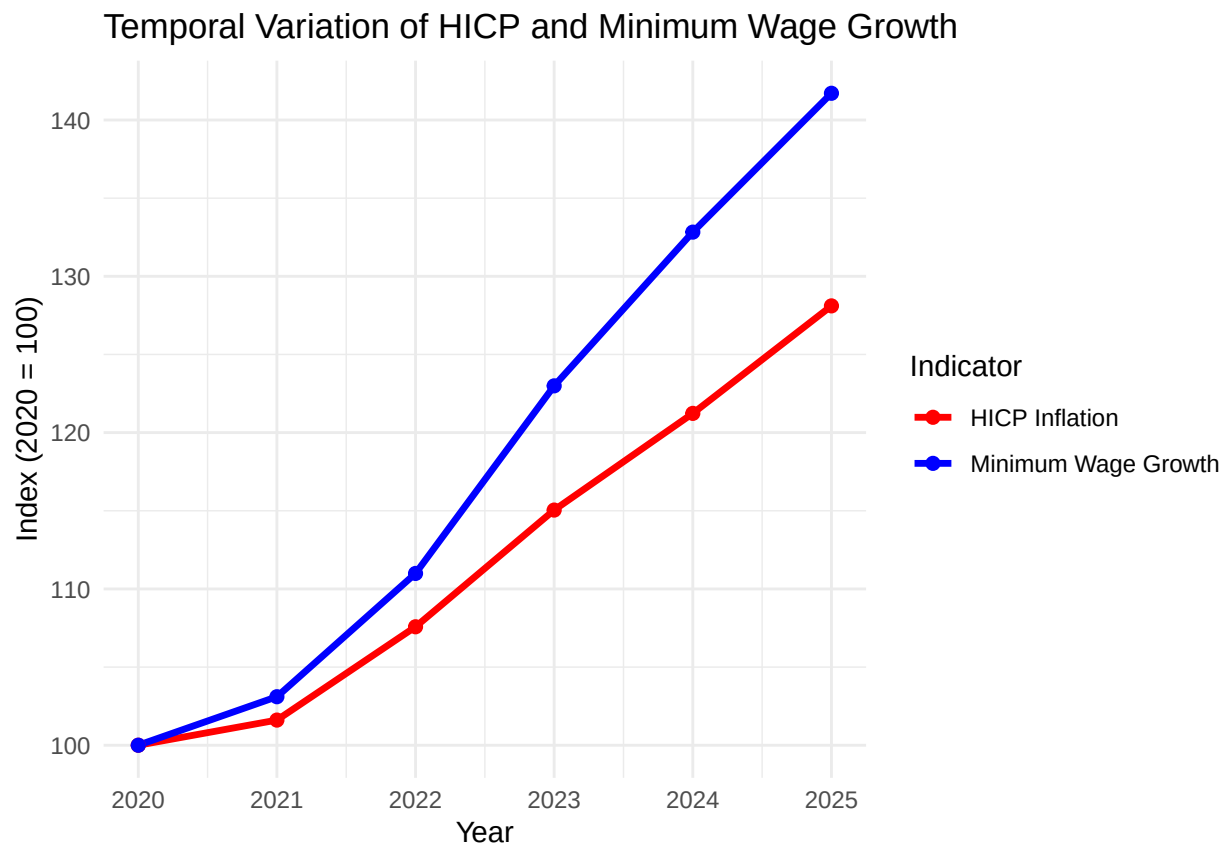
# =====
# Plot
# =====
ggplot(temporal_long,
  aes(x = year,
    y = value,
    colour = variable)) +

```

```

geom_line(linewidth = 1.2, na.rm = TRUE) +
geom_point(size = 2, na.rm = TRUE) +
labs(
  title = "Temporal Variation of HICP and Minimum Wage Growth",
  x = "Year",
  y = "Index (2020 = 100)",
  colour = "Indicator"
) +
scale_colour_manual(
  values = c(
    "avg_hicp" = "red",
    "avg_wage_growth" = "blue"
  ),
  labels = c(
    "HICP Inflation",
    "Minimum Wage Growth"
  )
) +
theme_minimal()

```

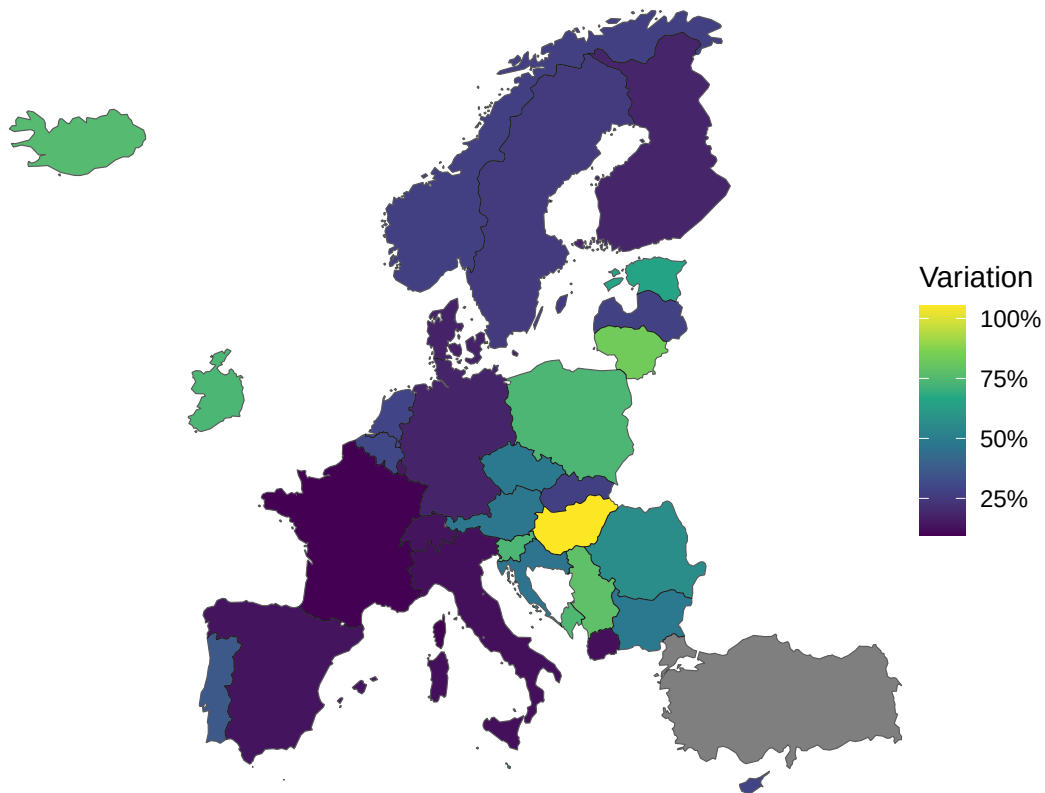


3.4 Visualize spatial variation

```
map_data <- left_join(europe, rent, join_by("NAME_ENGL" == "TIME")) %>%
  st_as_sf()

ggplot(map_data) +
  geom_sf(aes(fill = Var)) +
  coord_sf(
    xlim = c(-25, 45),
    ylim = c(34, 72),
    expand = FALSE
  ) +
  labs(title = "Map rent", fill = "Variation") +
  scale_fill_viridis_c(labels = scales::label_percent(accuracy = 1)) +
  theme_void()
```

Map rent



Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

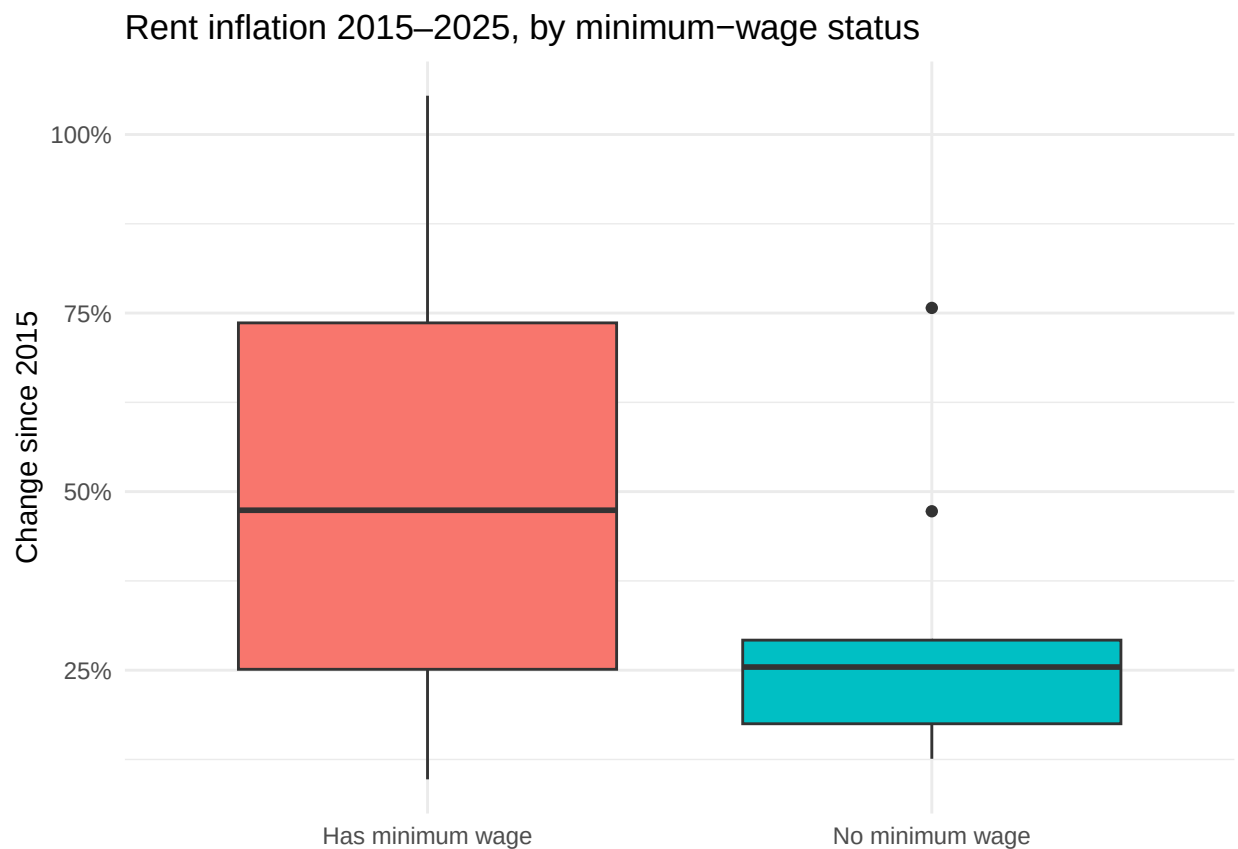
```
# Step 1: which countries have a minimum wage? (the ones that show up in the wage data)
wage_countries <- unique(wage_long$TIME)

# Step 2: label each country as being in one group or the other
```



```
subpop <- rent %>%
  mutate(min_wage_status = ifelse(TIME %in% wage_countries,
                                   "Has minimum wage", "No minimum wage"))

# Step 3: boxplot comparing rent inflation between the two groups
ggplot(subpop, aes(x = min_wage_status, y = Var, fill = min_wage_status)) +
  geom_boxplot() +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Rent inflation 2015–2025, by minimum-wage status",
       x = NULL, y = "Change since 2015") +
  theme_minimal() +
  theme(legend.position = "none")
```



Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

```
# Summarise average wage growth per country over the full period
wage_avg <- wage_long %>%
  group_by(TIME) %>%
  summarise(avg_wage_growth = mean(wage_growth, na.rm = TRUE))

# Join with rent variation
```

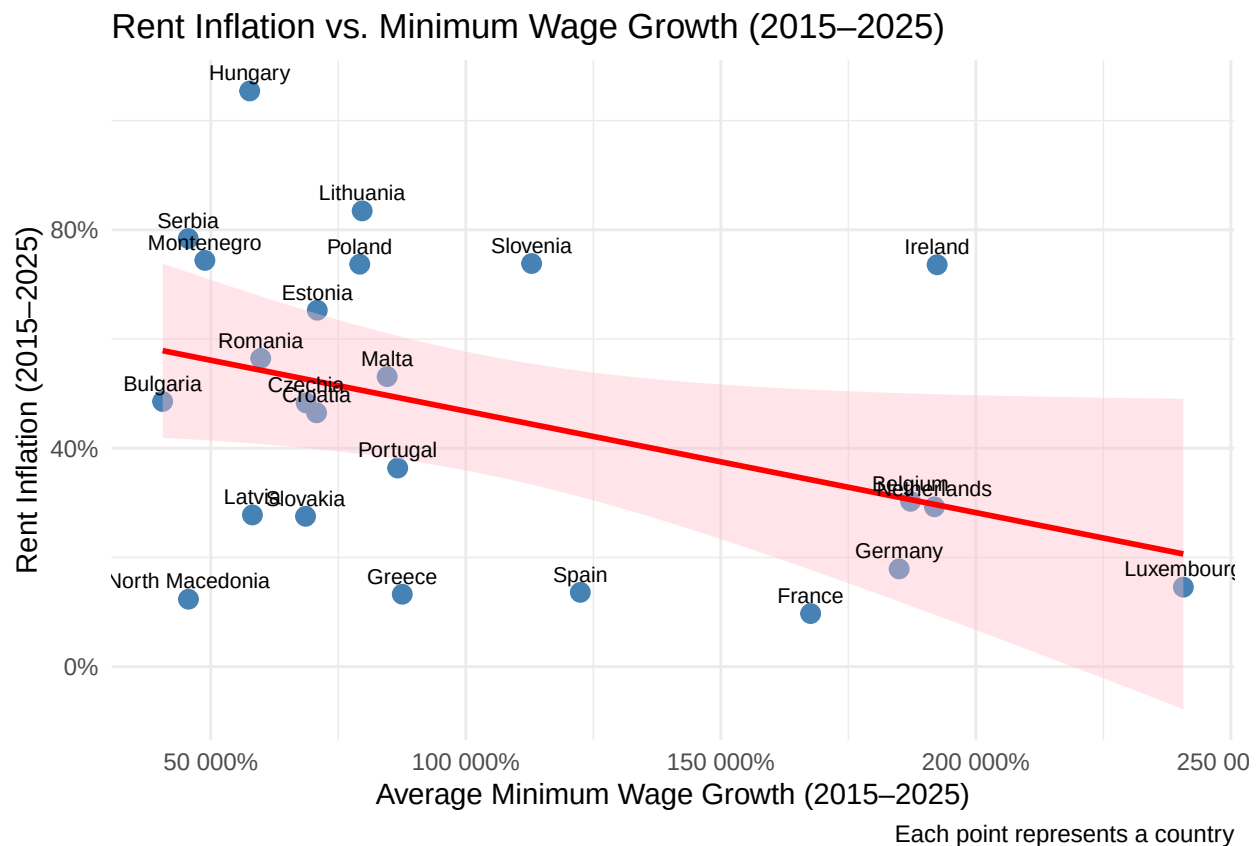
```

scatter_data <- left_join(
  rent %>% select(TIME, Var),
  wage_avg,
  by = "TIME"
) %>%
  filter(!is.na(Var), !is.na(avg_wage_growth))

# Scatter plot
ggplot(scatter_data, aes(x = avg_wage_growth, y = Var)) +
  geom_point(colour = "steelblue", size = 3) +
  geom_smooth(method = "lm", se = TRUE, colour = "red", fill = "pink") +
  geom_text(aes(label=TIME), vjust = -0.7, size = 2.8) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  scale_x_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(
    title = "Rent Inflation vs. Minimum Wage Growth (2015–2025)",
    x = "Average Minimum Wage Growth (2015–2025)",
    y = "Rent Inflation (2015–2025)",
    caption = "Each point represents a country"
  ) +
  theme_minimal()

```

'geom_smooth()' using formula = 'y ~ x'



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.